Bank Marketing Subscription Prediction

(Progress Review)

Team 12

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Outline

1. Progress Update
2. Introduction
3. Problem Setting
4. Related Work
5. Expected Methods
6. Experiments Setting
7. Experiments Results
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Outline

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Progress Update

- progress update in bullet points
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Background

- Marketing campaigns are crucial in the banking industry.
- **Term deposits** are an important product that provides stable funding for banks.
- Traditional marketing strategies are costly and often achieve low conversion rates.
- **Data-driven approaches** can improve targeting, reduce expenses, and enhance profitability.



Motivation

1. Traditional marketing is inefficient → few subscriptions
2. Data-driven strategy improves ROI
3. Compare multiple models to understand performance trade-offs between accuracy and interpretability.
4. Apply interpretable insights to support marketing strategy refinement and customer targeting.

Research Aims

1. Build a **predictive model** for term deposit subscription (Yes/No)
2. Explore **feature** importance for interpretability and marketing insights.
3. Compare algorithms (LogReg, Trees, RF, Boosting, DL)
4. Ensure robust generalization (stratified k-fold CV)
5. Benchmark vs Kaggle baselines & related works

Challenges

1. Class Imbalance Problem
    - There exists a significant imbalance between positive and negative classes.
    - To mitigate bias, we apply resampling techniques such as SMOTE on the training set only and evaluate the model using AUROC and F1-score to ensure balanced performance.
2. Overfitting Risk and Model Generalization
    - Complex models are prone to overfitting. We address this by applying regularization and early stopping during training.
    - Cross-validation is further used not as a prevention, but as a method to assess model robustness and generalization consistency.

Expected Contributions

1. Develop an accurate and interpretable prediction model
    - Balance performance and explainability to support marketing decisions.

2. Reveal key factors influencing customer subscription
    - Use SHAP analysis to provide actionable insights for targeted campaigns.

3. Compare classical and ensemble models
    - Analyze performance–interpretability trade-offs using AUC and F1 metrics.

Outline

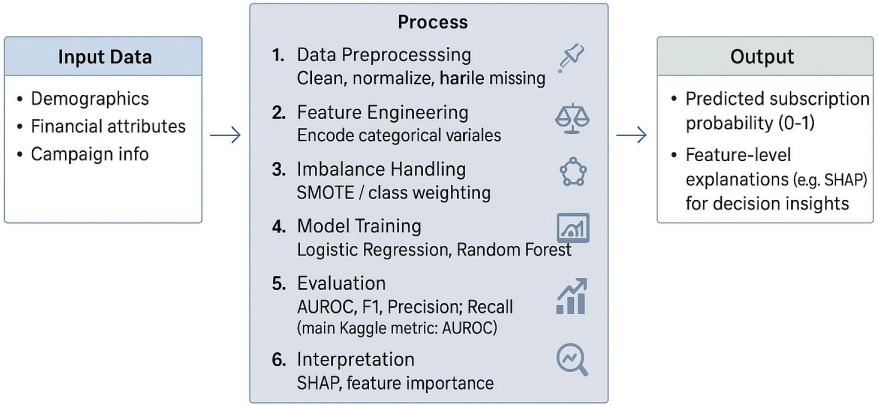
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Problem Description

1. Input
   a. Demographics, financial attributes, campaign info

2. Process:
   a. **Data Preprocessing**: Clean, normalize, handle missing values
   b. **Feature Engineering**: Encode categorical variables
   c. **Imbalance Handling**: SMOTE / class weighting
   d. **Model Training**: Logistic Regression, Random Forest, CatBoost
   e. **Evaluation**: AUROC, F1, Precision, Recall
   f. **Interpretation**: SHAP, feature importance

3. Output:
   a. Predicted subscription probability (0–1)
   b. Feature-level explanations for decision insights



Problem Description

Research Problem:

- This study aims to develop and evaluate interpretable machine learning models for predicting bank clients' subscription behavior.
- By addressing class imbalance and enhancing model explainability, the project seeks to improve targeting accuracy, reduce marketing costs, and provide actionable insights into key behavioral and demographic factors influencing client decisions.

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Selected Literature List

1. **Enhancing Customer Subscription Prediction in Bank Telemarketing Using Deep Learning and Ensemble Model [1]**
   - Source: IJRASET
   - Year: 2025
   - Authors: Dinesh Kumar Katakam
2. Predicting Customer Subscription in Bank Telemarketing Campaigns Using Ensemble Learning Models
   - **Source:** *Machine Learning with Applications*, Vol. 19, 2025
   - **Authors:** Michael Peter *et al.* (Nelson Mandela African Institution of Science and Technology)
   - Year: 2025
3. **Improving Imbalanced Data Classification in Bank Marketing Using Cluster PCA-Based Weighted Random Forest**
   - Source: *International Symposium on Pattern Recognition (ISPR 2024), CCIS 2303, Springer*
   - Year: 2025
   - Author: Dalia Atif
4. 4
5. *Unveiling the Power: A Comparative Analysis of Data Mining Tools through Decision Tree Classification on the Bank Marketing Dataset*
   - **Source:** *WSEAS Transactions on Computers, Vol. 23, 2024, pp. 95–105*
   - **Year:** 2024
   - **Authors:** *Elif Akkaya*, *Safiye Turgay*

Technical Side Summary

	Methodology	Performance	SOTA / Baseline	Insights / Pitfalls
1	Deep Learning (**RNN**, MLP), Ensemble (Bagging, Boosting, Stacking)	**Accuracy: 94.55%, F1-score: 94.01%**	**RNN = SOTA**	**RNN's ability to capture sequential patterns leads to superior performance** compared to ensemble methods.
2	**Ensemble** Learning (Bagging, Boosting, Stacking)	Acc 91.88%, AUC 0.9491	**Boosting model= Baseline SOTA = Stacking; Baseline = RF**	Ensemble methods **lack recall power** for minority classes Stacking best overall; bagging handles imbalance better
3	**Cluster PCA-Based Weighted Random Forest (CPWRF)**	Weighted-RF: Recall = 0.55, F1 = 0.58, Specificity = 0.96		Preserves original data distribution Reduces false positives and overfitting from synthetic data Sensitive to outliers with high PCA weights Requires hyperparameter tuning for clusters and PCA components
4	**CatBoost** + **SHAP**	Accuracy 90.91% AUC 93.82%	Comparison Method	**CatBoost** offered the best balance between **predictive power and stability**. **SHAP** provided clear feature importance, improving **interpretability and trust**
5	C4.5 Decision Tree on Bank Marketing dataset using 5 tools with k-fold CV	Orange achieved best accuracy (98.67%)	Baseline: C4.5; SOTA: Orange implementation	High interpretability but overfitting risk; tool settings affect results

Brief Introduction - 1

1. **Enhancing Customer Subscription Prediction in Bank Telemarketing Using Deep Learning and Ensemble Model [1]**
   - Authors: Dinesh Kumar Katakam
   - Source: IJRASET, 2025
   - Compares deep learning models (**RNN**, **MLP**
   - ensemble methods (**bagging**, **boosting**, **stack**
   - RNN is best

Model	Accuracy	Precision	Recall	F1 Score
RNN	**94.55%**	**89.54%**	**98.95%**	**94.01%**
MLP	93.64%	88.47%	98.07%	93.02%
Random Forest	90.65%	63.15%	39.13%	48.32%
Bagging	91.01%	61.84%	51.09%	55.95%
Gradient Boosting	90.17%	57.75%	44.56%	53.03%
AdaBoost	89.08%	56.90%	35.86%	44.00%
Stacking	90.53%	60.60%	43.48%	50.63%

Brief Introduction - 2

- Predicting Customer Subscription in Bank Telemarketing Campaigns Using Ensemble Learning Models
- **Source:** *Machine Learning with Applications*, Vol. 19, 2025
- Year: 2025
- **Authors:** Michael Peter *et al.* (Nelson Mandela African Institution of Science and Technology)
- M. Peter, H. Mofi, S. Likoko, J. Sabas, R. Mbura, and N. Mduma, "Predicting customer subscription in bank telemarketing campaigns using ensemble learning models," *Machine Learning with Applications*, vol. 19, p. 100618, 2025, doi: 10.1016/j.mlwa.2025.100618.

- **Input:**
  UCI Bank Marketing dataset (Portuguese bank telemarketing data, 41,188 samples).
  Features include demographics, campaign history, and macroeconomic indicators.
- **Process:**
  1. Data preprocessing — cleaning, categorical encoding, scaling, and imbalance handling (SMOTE).
  2. Implementation of **Bagging (Random Forest)**, **Boosting (Gradient Boosting, AdaBoost)**, and **Stacking** ensembles.
  3. Model optimization through grid search and stratified k-fold cross-validation.
- **Output:**
  Predict whether a customer subscribes to a term deposit (binary classification).

Brief Introduction - 2

Experimental Setting

- **Dataset:** UCI Bank Marketing (Moro *et al.*, 2014)
- **Tools:** Python (scikit-learn, imbalanced-learn)
- **Validation:** Stratified K-Fold Cross-Validation
- **Metrics:** Accuracy, Precision, Recall, F1-score, ROC-AUC, MCC, Cohen's Kappa

Experimental Results

- **Best overall model:** Stacking (Accuracy = 91.88%, ROC-AUC = 0.9491)
- **Best balance on imbalanced data:** Bagging Classifier (highest MCC and Kappa)
- **Insight:** Ensemble learning significantly improves prediction reliability for bank subscription tasks.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	MCC	Cohen's kappa
BM							
RF	0.9139	0.6619	0.4941	0.5658	0.9413	0.5294	0.5230
BC	0.9133	0.6356	0.5540	0.5920	0.9396	0.5621	0.5604
BOM							
GB	0.9185	0.6828	0.5273	0.5951	0.9476	0.5598	0.5567
ADB	0.9058	0.6916	0.3070	0.4252	0.9361	0.4748	0.4570
STCK							
ST	0.9188	0.6832	0.5305	0.5972	0.9491	0.5588	0.5549

Note: **BM**: Bagging Model, RF: Random Forest, BC: Bagging Classifier, **BOM**: Boosting Model, GB: Gradient Boosting, ADB: AdaBoost, **STCK**: Stacking Model, ST: Stacking.

Brief Introduction - 3

- **Improving Imbalanced Data Classification in Bank Marketing Using Cluster PCA-Based Weighted Random Forest**
- **Source:** *International Symposium on Pattern Recognition (ISPR 2024), Springer CCIS 2303 (2025)*
- **Author:** Dalia Atif (University of Tipaza, Algeria)
- **Year:** 2025
- ATIF, D. (2025). Improving Imbalanced Data Classification in Bank Marketing Using Cluster PCA-Based Weighted Random Forest. In: Bennour, A., Bouridane, A., Almaadeed, S., Bouaziz, B., Edirisinghe, E. (eds) Intelligent Systems and Pattern Recognition. ISPR 2024. Communications in Computer and Information Science, vol 2303. Springer, Cham. https://doi.org/10.1007/978-3-031-82150-9_23

- **Input:** UCI Bank Marketing Dataset (45,211 records, 16 input features, binary target: client subscribed to term deposit or not). Highly imbalanced with 11.7% positive samples.
- **Process:** ① Encode categorical variables → ② Split data 80:20 (train/test) → ③ Apply clustering (K-means) on minority class → ④ Apply PCA within each cluster to compute contribution weights → ⑤ Train a **Weighted Random Forest** using these instance weights.
- **Output:** Improved classification model achieving higher recall and balanced precision without distorting data distribution.

Brief Introduction - 3

- **Dataset:** UCI Bank Marketing Dataset (Portugal, 2008–2010), 45,211 samples, 16 features.
- **Preprocessing:** Label encoding for categorical data; training/testing split (80:20).
- **Hyperparameters**: Clusters = 15, Principal Components = 5, Trees = 2000.
- **Comparison Models:** RF, LR, SVM with and without SMOTE / K-SMOTE / ADASYN; Proposed Weighted-RF.
- **Repetitions:** 30 runs to ensure robustness (mean performance reported).

- The proposed **Cluster PCA-based Weighted RF** achieves the **best trade-off** between Recall and Precision (F1=0.58) while maintaining high Specificity (0.96).
  It avoids the overfitting and false positives common in oversampling-based methods.

Table 3. Performance scores for different imbalance treatments

Method	Recall	Precision	F1-score	Specificity	Accuracy
RF	0.34	0.64	0.45	0.97	0.90
WEIGHTED-RF	0.55	0.61	0.58	0.96	0.91
SMOTE+RF	0.66	0.47	0.55	0.89	0.87
K-SMOTE+RF	0.46	0.58	0.51	0.95	0.89
ADASYN+RF	0.68	0.49	0.57	0.89	0.88
SMOTE+LR	0.69	0.32	0.44	0.80	0.79
K-SMOTE+LR	0.45	0.42	0.44	0.92	0.86
ADASYN+LR	0.74	0.30	0.45	0.79	0.78
SMOTE+SVM	0.70	0.30	0.42	0.79	0.77
K-SMOTE+SVM	0.34	0.40	0.37	0.93	0.86
ADASYN+SVM	0.73	0.29	0.41	0.75	0.75

Brief Introduction - 4

4. **Enhancing Bank Term Deposit Predictions: A Machine Learning Approach with CatBoost and SHAP. Applied and Computational Engineering [4]**
    - Authors: Qiwanlun Yu et al.
    - Source: Applied and Computational Engineering, January 2025
    - Summary: This paper applies Categorical Boosting (**CatBoost) with Shapley Additive Explanations (SHAP)** to improve the **accuracy** and **interpretability** of bank term deposit subscription predictions.
    - **Enhancing Bank Term Deposit Predictions: A Machine Learning Approach with CatBoost and SHAP**
      Proc. of the 5th International Conference on Signal Processing and Machine Learning, 2024.
      Author: Qiwanlun Yu *(Zhongnan Univ. of Economics and Law, China)*
      DOI: 10.54254/2755-2721/120/2025.19485

Input–Process–Output

InputBank Marketing dataset (Kaggle, 15 features + 1 target).

Process Compared 9 ML models (DT, Bagging, RF, ET, AdaBoost, GBDT, XGBoost, LightGBM, CatBoost). Performed preprocessing (scaling, one-hot encoding) and hyperparameter tuning with grid search. Applied SHAP for interpretability analysis.

OutputOptimized CatBoost model (Accuracy = 90.91%, AUC = 93.82%) with global + local explainability via SHAP.

Brief Introduction - 5

- *Unveiling the Power: A Comparative Analysis of Data Mining Tools through Decision Tree Classification on the Bank Marketing Dataset*
- **Source:** *WSEAS Transactions on Computers, Vol. 23, 2024, pp. 95–105*
- **Authors:** *Elif Akkaya*, *Safiye Turgay*
- **Year:** 2024
- E. Akkaya and S. Turgay, “Unveiling the Power: A Comparative Analysis of Data Mining Tools through Decision Tree Classification on the Bank Marketing Dataset,” *WSEAS Transactions on Computers*, vol. 23, pp. 95–105, May 2024.
- **Input:**
  - UCI Bank Marketing dataset (45,211 instances, 17 features) , Target variable: whether the customer subscribes to a term deposit (y)
- **Process:**
  - Data cleaning: handling missing values, normalization, and encoding
  - Model training using the **C4.5 Decision Tree** algorithm
  - Comparison across five data mining tools: **Weka, RapidMiner, KNIME, Orange, and Tanagra**
  - Evaluation through **k-fold cross-validation**
  - Performance assessed using **Accuracy, Precision, Recall, F1-score, and AUC-ROC** metrics
- **Output:**
  - Classification results and confusion matrices for each tool
  - Visualization of decision tree structures
  - Statistical comparison using **paired t-test** to determine performance significance

Brief Introduction - 5

Experimental Setting

- **Dataset:** UCI Bank Marketing dataset (Portuguese banking institution, direct marketing campaigns)
- **Algorithm:** C4.5 Decision Tree
- **Tools Compared:** Weka, RapidMiner, KNIME, Orange, Tanagra
- **Validation Strategy:** k-fold Cross Validation
- **Metrics:** Accuracy, Precision, Recall, F1-score, AUC-ROC

Experimental Results

- **Orange** achieved the **highest accuracy (98.67%)**, followed by **RapidMiner**, **Weka**, **KNIME**, and **Tanagra**.

Interpretation

- Orange outperformed others due to its **user-friendly interface**, **strong visualization**, and **efficient feature processing**.
- KNIME and Weka demonstrated **better stability and consistency**.
- The Decision Tree model showed signs of **overfitting**, suggesting that **pruning or ensemble techniques** could improve generalization.

Conclusion

- **Recurrent Neural Networks (RNN)** currently represent the **SOTA**
  - Due to model **sequential and temporal dependencies**
- Ensemble methods lack recall power
  - For minority classes
- Bayesian methods provide **interpretable models**
  - For imbalanced datasets with explanatory power.
- **CatBoost with SHAP** improving both model performance and interpretability.
- Insights:
  - Explore advanced sequential models (**LSTM**/**GRU**)
  - **real-time deployment** model explainability tools (**SHAP**, **LIME**)

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Dataset Overview

1. Dataset:
    - Kaggle Playground S5E8 (Bank Marketing Subscription)
    - https://www.kaggle.com/competitions/playground-series-s5e8/overview
2. Source:
    - Synthetic version of UCI Bank Marketing (Moro et al., 2014)
3. Size: 750k train, 250k test
4. Features:
    - Categorical: job, marital, education, housing, loan, contact type, etc.
    - Numerical: age, balance, duration, previous contacts, etc.
5. Target: subscribed (binary Yes/No)

Key Observations

- No missing values, all columns clean and consistent.
- Highly imbalanced target: positive (≈11%) vs. negative (≈89%).
- Strongest signal: duration positively correlated with subscription.
- Most categorical variables weakly correlated, implying non-linear patterns.

Preprocessing Steps

- Convert categorical variables to category
  - For efficient memory and encoding
  - Solve challenge: Model Generalization

```
# 1) cast to category
for df in (train, test, orig):
    for c in CATS:
        df[c] = df[c].astype('category')
```

Preprocessing Steps

- Normalize / log-transform skewed numeric variables
  - balance_log, duration_sin/cos, pdays_sin
  - Stabilize distribution
  - Solve challenge: Overfitting Risk and Model Generalization

```python
def add_num_transforms(df):
    # log with sign for balance
    df['_balance_log'] = (np.sign(df['balance']) * np.log1p(np.abs(df['balance']))).astype('float32')
    # cyclical for duration (例：以 540s 為週期，可依資料調整)
    df['_duration_sin'] = np.sin(2*np.pi * df['duration'] / 540).astype('float32')
    df['_duration_cos'] = np.cos(2*np.pi * df['duration'] / 540).astype('float32')
    # cyclical for pdays (例：以 365 天對應週期)
    df['_pdays_sin'] = np.sin(2*np.pi * np.clip(df['pdays'], -1, 365) / 365).astype('float32')
    return df
```

Preprocessing Steps

- Create interaction features (numeric bigrams)
  - Capture non-linear effects between variables.
  - Solve challenge: Weakly Correlated

```
from itertools import combinations

NUMS_ENH = NUMS + ['_balance_log','_duration_sin','_duration_cos','_pdays_sin']

def add_numeric_bigrams(df, cols, cap=32):
    pairs = list(combinations(cols, 2))[:cap]
    for a, b in pairs:
        df[f'BIG_{a}__{b}'] = (df[a].astype('float32') * df[b].astype('float32'))
    return df
```

Preprocessing Steps

- Apply Target Encoding (TE) and Count Encoding (CE)
  - Uing both competition and original UCI dataset
  - Enriched categorical representations.
  - Solve challenge: Class Imbalance

```
# Count Encoding from UCI
for c in CATS:
    vc = orig[c].value_counts()
    train[f'CE_ORIG_{c}'] = train[c].map(vc).fillna(0).astype('float32')
    test[f'CE_ORIG_{c}']  = test[c].map(vc).fillna(0).astype('float32')
```

Preprocessing Steps

- Apply Target Encoding (TE) and Count Encoding (CE)
  - Uing both competition and original UCI dataset
  - Enriched categorical representations.
  - Solve challenge: Class Imbalance

```
from cuml.preprocessing import TargetEncoder

def apply_te_in_fold(X_tr, y_tr, X_va, X_te, cats, seed=42):
    # 只在當前 fold 的訓練資料上 fit；val/test 僅 transform（避免洩漏）
    for c in cats:
        enc = TargetEncoder(n_folds=10, smooth=0, seed=seed,
split_method='random', stat='mean')
        X_tr[c] = enc.fit_transform(X_tr[c], y_tr)
        X_va[c] = enc.transform(X_va[c])
        X_te[c] = enc.transform(X_te[c])
    return X_tr, X_va, X_te
```

Preprocessing Steps

- Ensure no data leakage
  - Transformations using y (e.g., TE, SMOTE) are fitted on training folds only
  - External statistics (e.g., CE from orig) are computed offline without target labels.
  - Solve challenge: Overfitting & Generalization

Modeling Strategy

- Explore both tree-based and deep tabular architectures
  - Ensure diversity and robustness.
- Use Stratified 5-fold Cross-Validation for consistent evaluation across all models.
- Optimize hyperparameters with early stopping.
  - learning rate, embedding dim, dropout, epochs

Models for Feature-Weight Analysis

Model Type	Primary Method	Notes
Boosting / Tree Models (XGBoost, LightGBM, CatBoost, Random Forest)	**Gain-based importance** for initial ranking,followed by **Permutation Importance** for verification	Provides strong baseline; balances predictive power and interpretability.
Deep Tabular Models (TabM, FTTransformer, TabR, GRN, CNN, RealMLP, DeepTables)	**Permutation Importance (AUC drop)**	Unified evaluation metric across all architectures for fair comparison.
Ensemble Layer (Blender) (CatBoost meta-model)	**CatBoost feature importance on base-model scores**	Quantifies each base model's contribution to the final ensemble.

Implementation Plan

- Begin with TabM (TabMini architecture) as the first case study.
- Apply Permutation Importance to analyze the influence of each feature group (TE, CE, cyclical, numerical).
- Extend the same process to other models (XGB, LGBM, CatBoost, etc.).

Evaluation Metrics

Metric	Description	Purpose
Accuracy	Overall percentage of correct predictions.	General baseline indicator.
Precision	Proportion of predicted positives that are true positives.	Measures false-positive control.
Recall (Sensitivity)	Proportion of true positives that are correctly detected.	Important for minority class detection.
F1-score	Harmonic mean of Precision and Recall.	Balances precision–recall trade-off for imbalanced data.
AUROC	Area under the ROC curve.	Primary metric for Kaggle; robust to class imbalance.

Outline

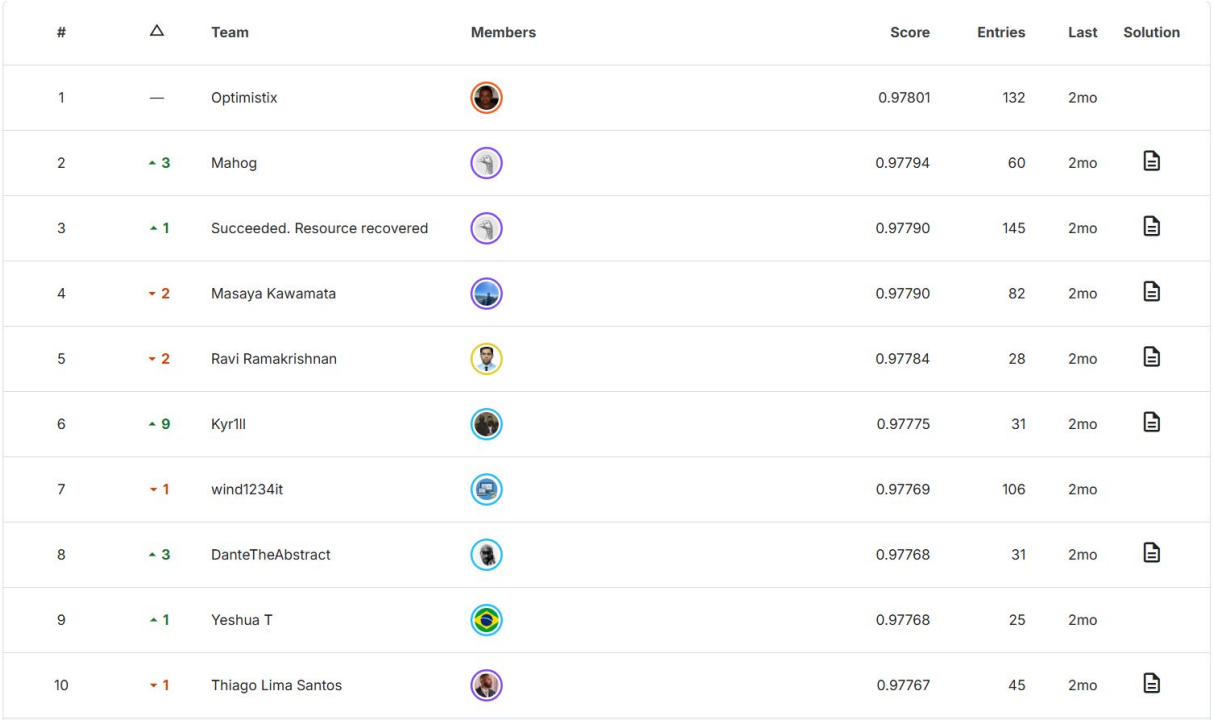
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Data Description and Characteristics

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#	△	Team	Members	Score	Entries	Last	Solution
1	—	Optimistix		0.97801	132	2mo	
2	▲ 3	Mahog		0.97794	60	2mo	
3	▲ 1	Succeeded. Resource recovered		0.97790	145	2mo	
4	▼ 2	Masaya Kawamata		0.97790	82	2mo	
5	▼ 2	Ravi Ramakrishnan		0.97784	28	2mo	
6	▲ 9	Kyr1ll		0.97775	31	2mo	
7	▼ 1	wind1234it		0.97769	106	2mo	
8	▲ 3	DanteTheAbstract		0.97768	31	2mo	
9	▲ 1	Yeshua T		0.97768	25	2mo	
10	▼ 1	Thiago Lima Santos		0.97767	45	2mo	

Dataset Checklist

1. Domain:
   - ✓ **Finance & Economics**
2. Source:
   - ✓ **Public** Kaggle competition (leaderboard & baselines available)
3. Size:
   - ✓ **1M** samples (≫ 5k minimum)
4. Features:
   - ✓ Clear semantics (age, job, contact, etc.) → interpretable
5. Task:
   - ✓ **Binary classification** (Yes/No subscription)

Basic Exploratory Data Analysis (EDA)

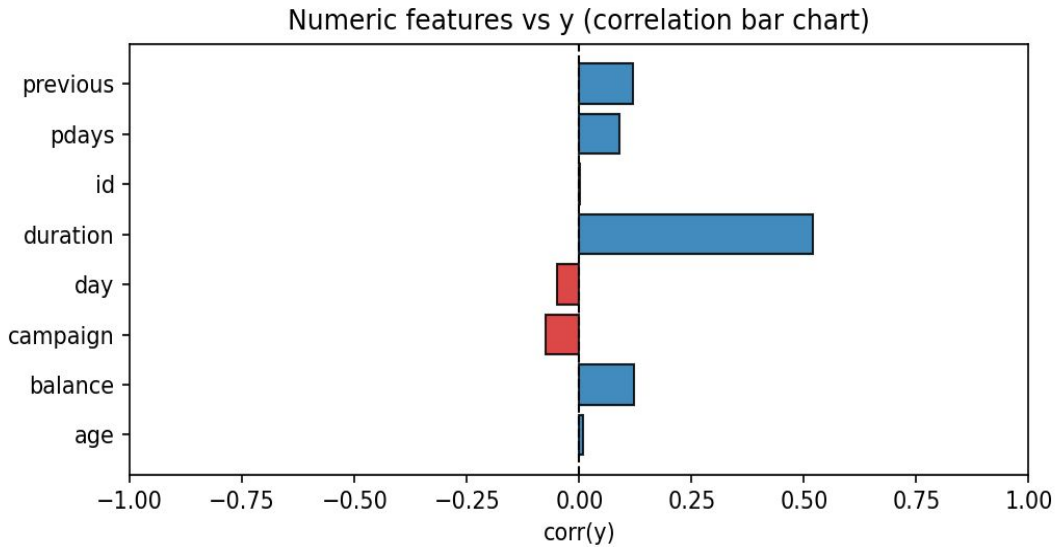
Variable	Description	Type / Values
age	Client’s age	Numeric
balance	Average yearly balance (EUR)	Numeric
day	Last contact day of month	Numeric (1–31)
duration	Last contact duration (seconds)	Numeric
campaign	Number of contacts in this campaign	Numeric
pdays	Days since last contact (-1 = never contacted)	Numeric
previous	Number of contacts before this campaign	Numeric
y	Subscribed to term deposit? (target)	Binary: yes, no

Basic Exploratory Data Analysis (EDA)

Variable	Description	Type / Values
job	Type of job	Categorical: admin., blue-collar, entrepreneur, etc.
marital	Marital status	Categorical: married, single, divorced
education	Education level	Categorical: primary, secondary, tertiary, unknown
default	Has credit in default?	Categorical: yes, no
housing	Has a housing loan?	Categorical: yes, no
loan	Has a personal loan?	Categorical: yes, no
contact	Contact communication type	Categorical: unknown, telephone, cellular
month	Last contact month	Categorical: jan, feb, …, dec
poutcome	Outcome of previous campaign	Categorical: unknown, other, failure, success

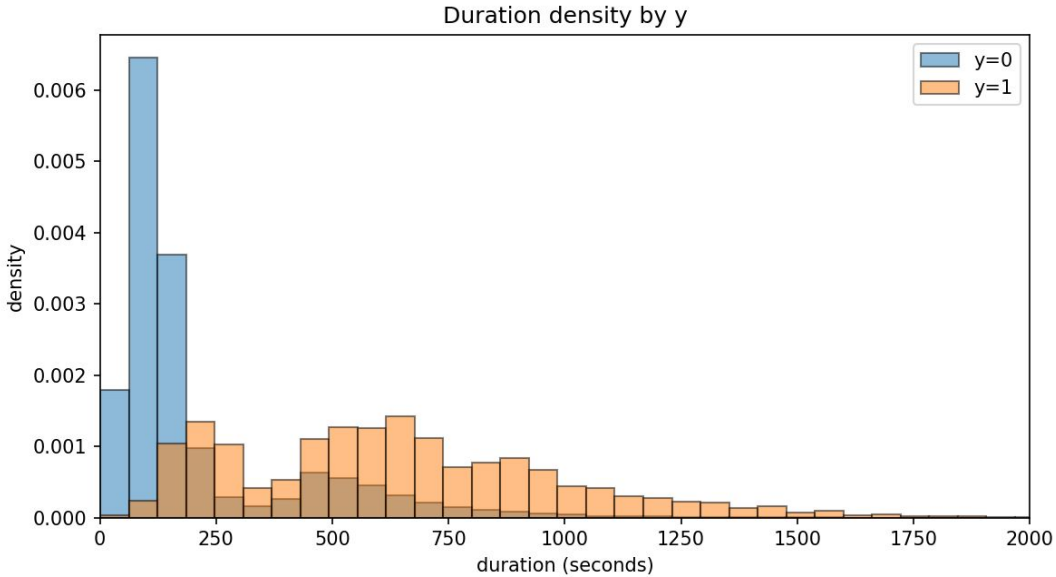
Basic Exploratory Data Analysis (EDA)

1. missing value rate: 0%
2. numeric vs y correlation analysis
   - duration is strongest signal



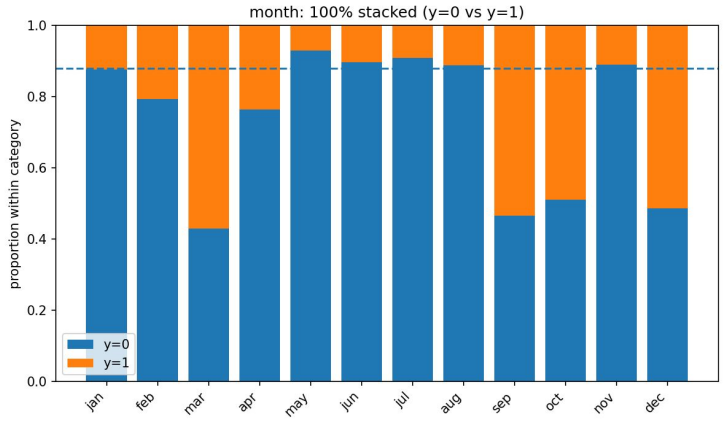
Basic Exploratory Data Analysis (EDA)

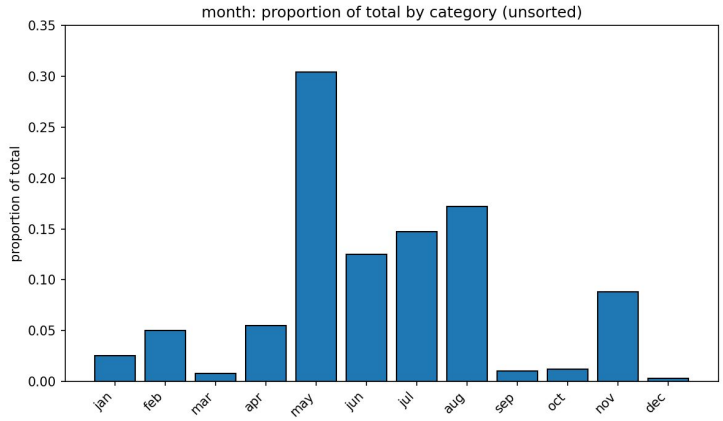
- numeric vs y correlation analysis (duration)
  - Interested clients talk longer



Basic Exploratory Data Analysis (EDA)

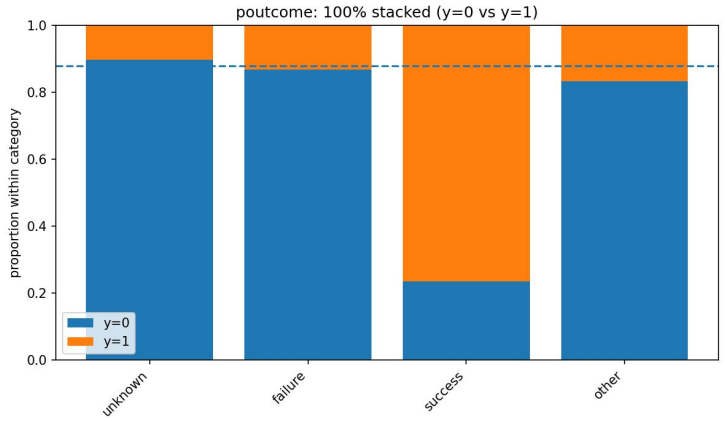
- Analyze non-numeric data (month)
  - In Mar, Sep, Oct, Dec, the proportion of true is high
  - However, their overall share of data is relatively low

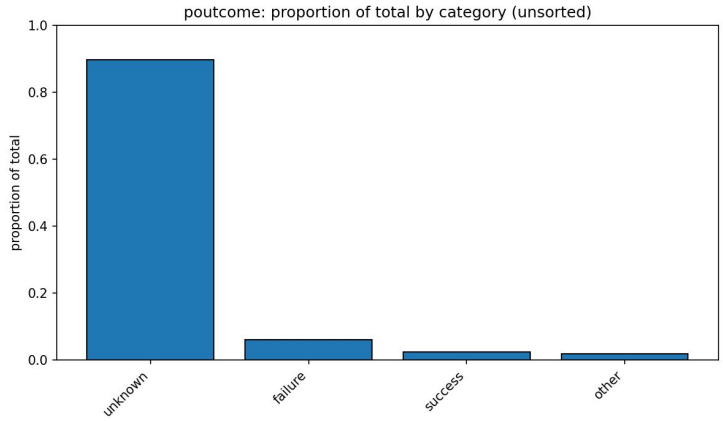




Basic Exploratory Data Analysis (EDA)

- Analyze non-numeric data (poutcome)
  - If the client had a previous successful outcome,
  - The probability of success is much higher
  - However, this category has very few samples





Basic Exploratory Data Analysis (EDA)

- Summary:
  - No missing data
  - Most categories show no clear correlation with y
  - Correlated categories cover only a small portion of data

Basic Exploratory Data Analysis (EDA)

- Dataset clean and usable → feasible for modeling
- Low correlation → signals are non-linear patterns
- Plan:
    a. Create interaction & composite features
    b. Use ensemble models for hidden relationships
    c. Apply SHAP for interpretability

Synthetic vs. Original Dataset

- Purpose:
  - To expand the training dataset and improve model generalization.
- Difference:
  - Generated by a deep model, so feature distributions are similar but not identical.
  - May alter rare patterns or correlations among variables.
- Potential Distortion:
  - Synthetic process may introduce bias or loss of realism, especially in edge cases.

Evaluation Strategy

- Use Stratified 5-Fold CV
- Evaluate each fold independently and compute OOF (Out-of-Fold) scores
- Compare cross-model results (TabM, XGB, LGBM, etc.) using the same metrics.
- Main evaluation metrics: AUROC (primary) and F1-score (secondary).
  - Solve Challenge: Class Imbalance

Outline

1. Progress Update
2. Introduction
3. Problem Setting
4. Related Work
5. Expected Methods
6. Experiments Setting
7. Experiments Results (running)
8. Next Steps

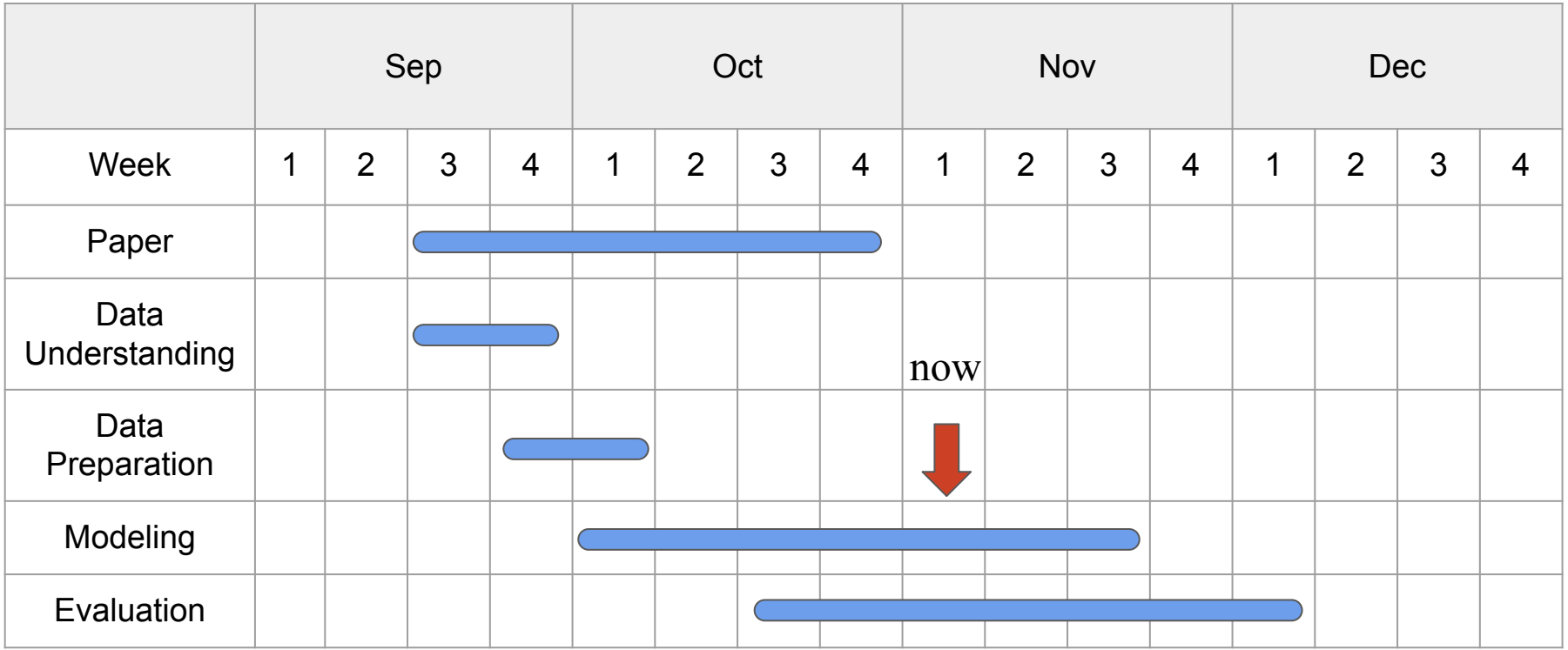
Outline

1. Progress Update
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6. Experiments Setting
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8. Next Steps

Implementation Plan

- Begin with TabM (TabMini architecture) as the first case study.
- Apply Permutation Importance to analyze the influence of each feature group (TE, CE, cyclical, numerical).
- Extend the same process to other models (XGB, LGBM, CatBoost, etc.).

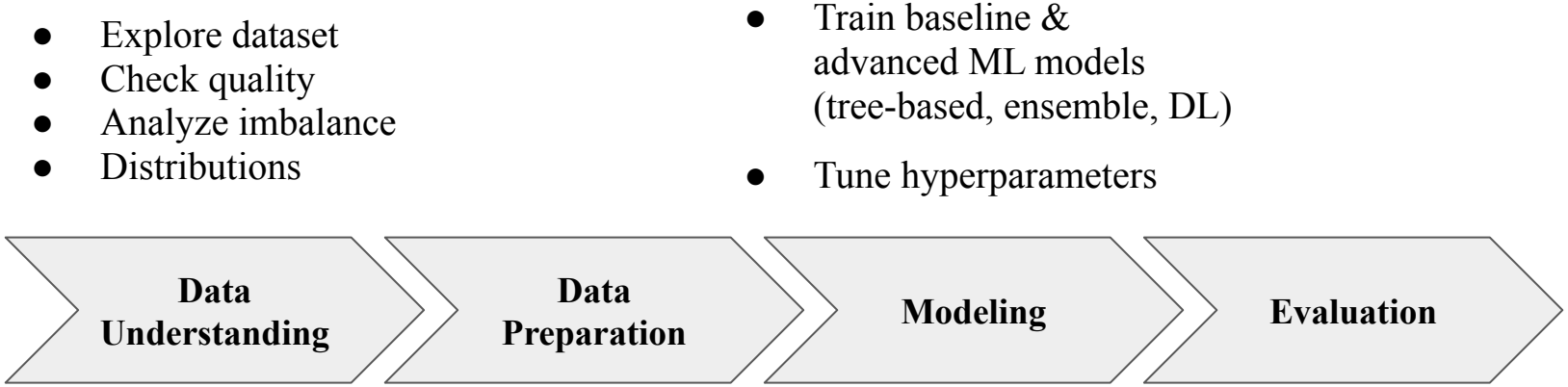
Next Steps - Schedule



Environment

1. OS
    - Linux / Windows
2. Programming Language
    - Python
3. Tools
    - VSCode
4. Libraries
    - pytorch, pandas, numpy, dgl

Analysis Workflow



- Explore dataset
- Check quality
- Analyze imbalance
- Distributions

- Clean & preprocess data (encoding, scaling)
- Handle class imbalance – apply SMOTE on training set only
- Feature engineering & selection

- Train baseline & advanced ML models (tree-based, ensemble, DL)
- Tune hyperparameters

- Use metrics (Accuracy, AUROC, F1, PR-AUC)
- Apply stratified k-fold CV & hold-out test

Reference

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